Introduction to Huawei Cloud ModelArts

Foreword

- ModelArts is a one-stop AI development platform. It provides data preprocessing, semi-automated data labeling, large-scale distributed training, automated model generation, and model deployment on devices, edge devices, and the cloud. It helps you develop and deploy models quickly and manage the AI development lifecycle for machine learning and deep learning.
- This chapter describes the main functions of Huawei ModelArts, helping you better understand and use it.

Objectives

- Upon completion of this course, you will understand:
  - Basic concepts of ModelArts.
  - Functions and usage of ModelArts.

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Contents

1. **ModelArts Overview**
2. ModelArts Functions

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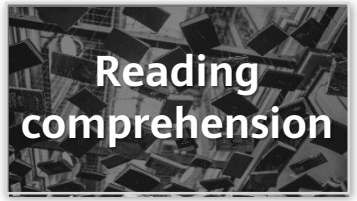
From AI+ to +AI : AI Explores Industry Best Practices

AI+

Exploring AI capabilities

EfficientNet model accuracy: 98.7% for the **top 5 labels**, surpassing the human-level accuracy of 96%

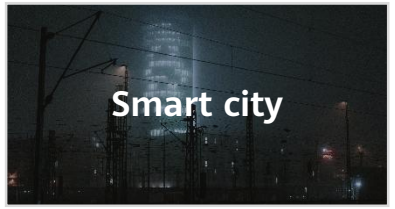
RNN-T model accuracy: 96.8%, surpassing the human-level accuracy of 94.17%



BERT model accuracy: 90.9%, surpassing the human-level accuracy of 82%

+AI

AI is empowering enterprises' core production systems



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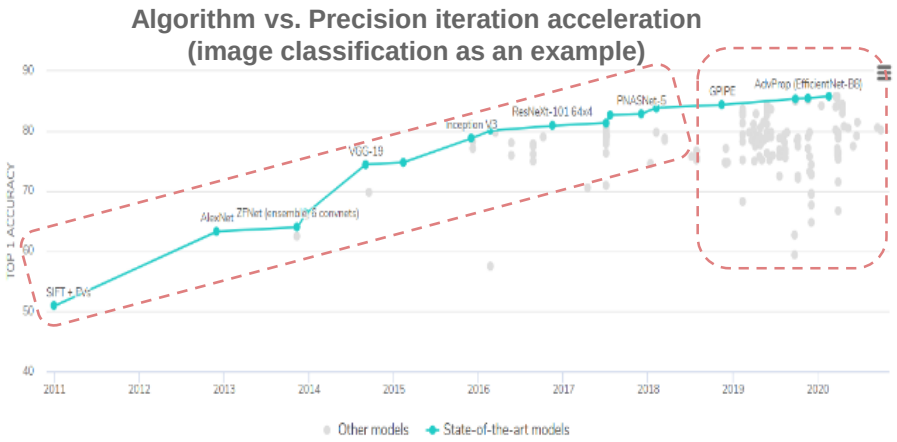
A Long Way to Go from the Thought of "AI + Industry" to Product-based Applications

AI development difficulties

Fast algorithm update; difficult algorithm selection and tuning

50%
Annual growth rate of global papers on AI

100+
Annual algorithm iteration in a single domain



AI implementation challenges

Difficult data acquisition

Difficult industry knowledge distillation

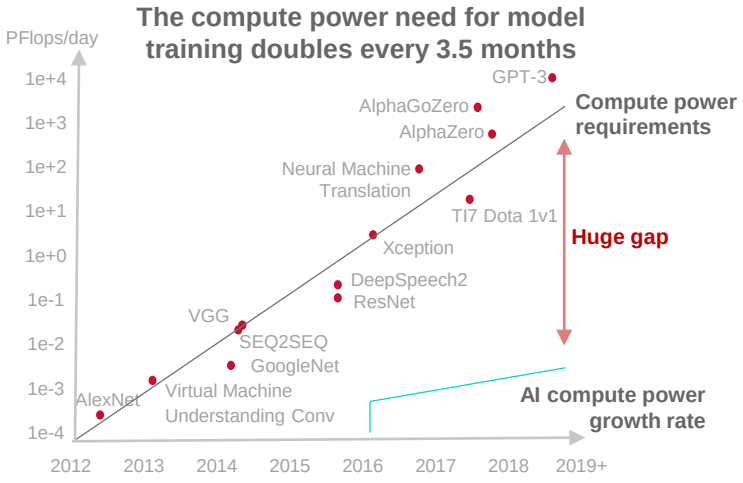
Difficult knowledge computing

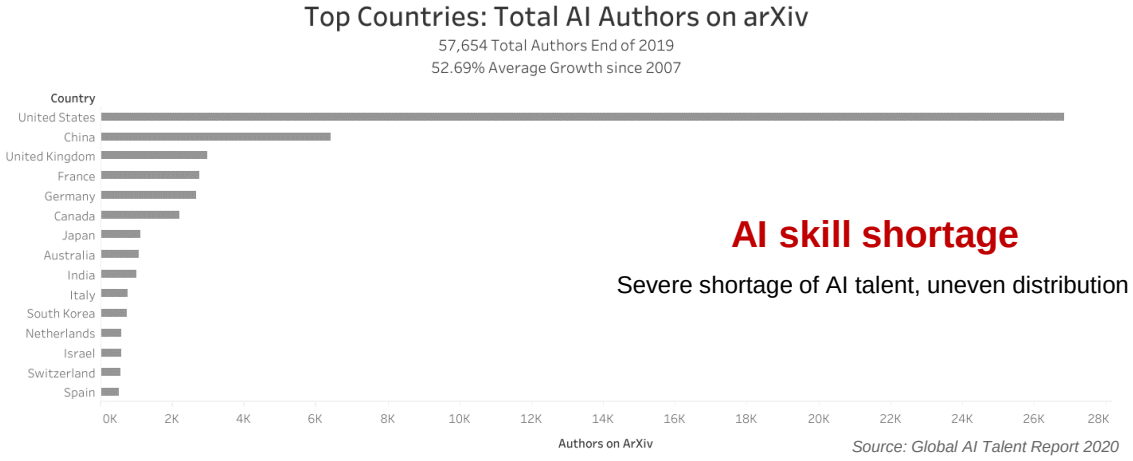
AI compute power scarcity

Expensive AI model training

GPT-3
US$4.6 million

BERT
US$15,000 on average





AI skill shortage

Severe shortage of AI talent, uneven distribution

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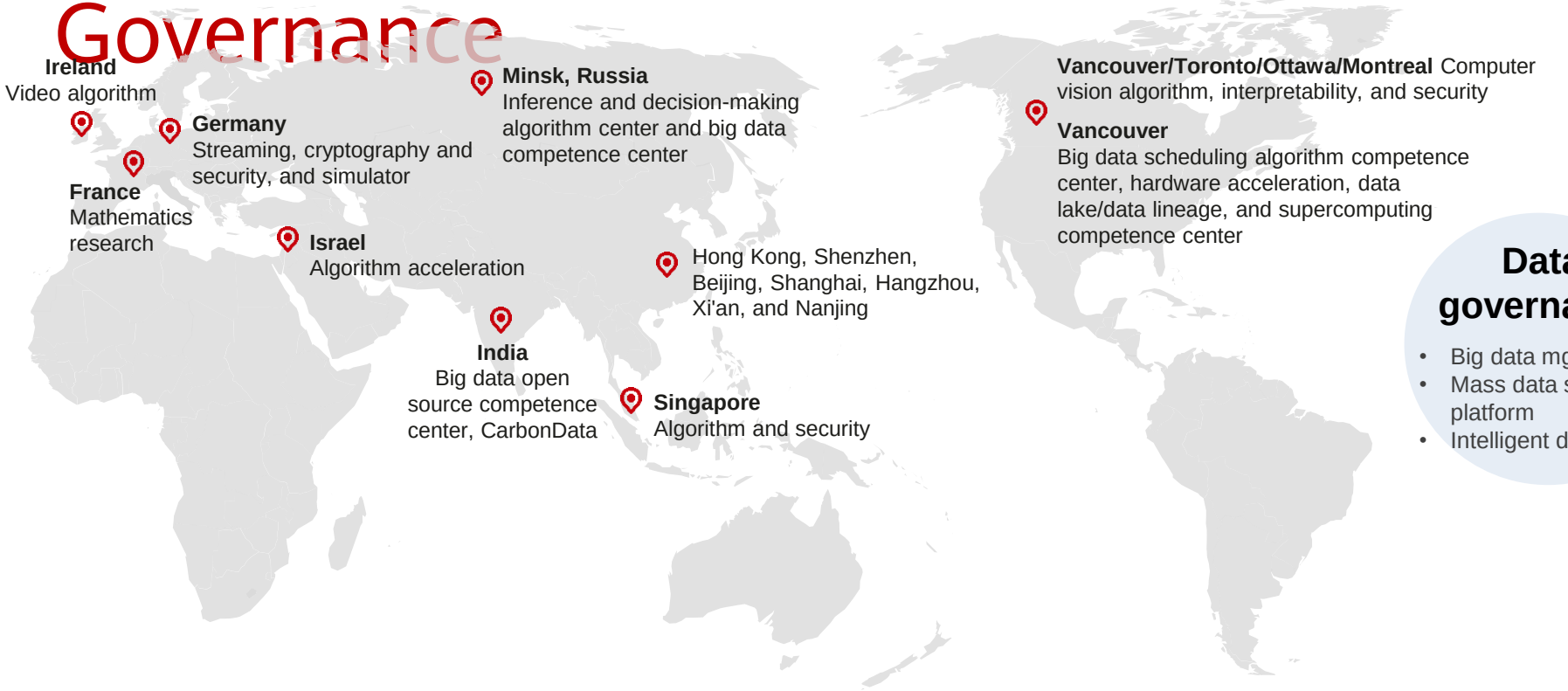
Global AI Research Deployment: Building Core Capabilities Around Computing, Algorithm, and Data Governance



Huawei AI

Algorithm
- Deep learning
- Data mining
- Knowledge graph
- Cognitive computing

Data governance
- Big data mgmt platform
- Mass data storage platform
- Intelligent data lake

Compute power
- Ascend processor
- Kunpeng processor

AI platform/ Algorithm framework
- ModelArts
- MindSpore

No. 1	10%+	15+	85+	5,000+
Ranking by patents	Percentage of PhDs	R&D centers in 4 continents	Cooperation with global organizations	R&D engineers

MindSpore, APACHE CarbonData

Top community contributor

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Building Powerful AI Technologies for ModelArts by Continuous Innovation and Research

No. 1 on ImageNet-1000

The highest accuracy of **85.8%** on the industry's most widely used, large-scale image classification dataset

No.1 in WebVision

82.97% accuracy in image classification with only slightly labeled WebVision dataset

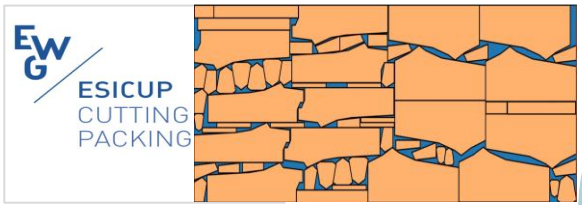
Intelligent perception

No.1 on MS COCO

56.8% single-model accuracy and **58.8%** multi-model accuracy on MS-COCO, one of the most widely used large-scale object detection datasets

No. 1 on NuScenes

64.2% accuracy based on multi-modal data convergence on one of the most widely used large-scale 3D object detection datasets



No. 1 on ESICUP

No. 1 utilization of **81.25%** on ESICUP, the most widely used public dataset on cutting and packing, by leveraging Huawei's proprietary decision optimization algorithm

Decision-making support

Huawei Cloud EI

Intelligent cognition

WSDM No.1
Knowledge graph and data mining

NLPCC No.1
Built-in language models

CCKS No.1
Event extraction for finance

CCF BDCI No.1
Entity-level sentiment analysis

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ModelArts: Ascend-based Full-Stack AI Platform

Fully controllable AI technology stack

Intelligent application	Scientific Research	Transportation	Education	Manufacturing	Finance	Healthcare	City

Full-stack AI

Software platform

ModelArts, full-process AI enablement platform

Algorithm development environment	ExeML	Pangu models	Industry datasets
Data labeling	Distributed debugging	AI scientific computing	Built-in Ascend algorithms
Data management	AI training	AI inference	AI asset marketplace

Controllable in-house chip enablement tool chain

Deep learning computing framework MindSpore	Ascend operator development tool MindStudio

Ascend heterogeneous computing driver CANN

Hardware infrastructure

In-house chips with intellectual property

Kunpeng (general compute power)	Ascend NPUs (AI compute power)

Infrastructure construction

Planning and design	Civil works	Electricity	Cooling, ventilation, etc

Advantages

Future-proof ecosystem → **AI-enabled industries**

G42

100+ AI models

AI-assisted drug screening

1000+ algorithms/models

AI-enabled genome analysis

Distributed optimization → **Large-scale AI**

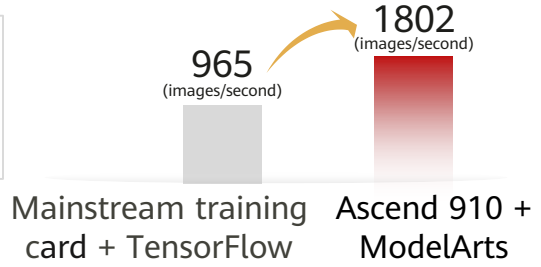
IO500 AIPerf — Double first

World's first EB-class compute cluster

- A single task supports ≥ **4,096** cards
- A single cluster supports ≥ **100,000** cards

Hardware-software synergy → **High-performance AI**

ResNet-50 Same computing power



800+ projects & practices in 10+ industries, creating new value with AI software

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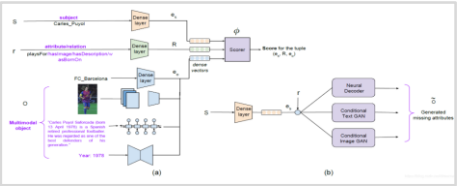
ModelArts: Core Technologies for AI Adoption

Industry expression

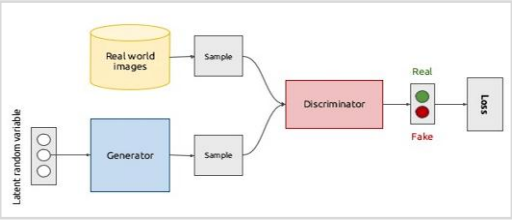
Reduce $\|f_{T_I \to T_A}\|$

Multimodal knowledge representation

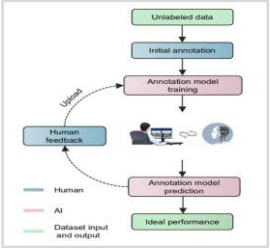


Knowledge graph

Adversarial learning and sample generation



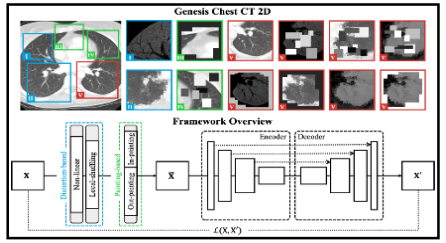
Auto labeling



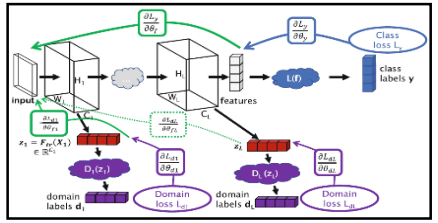
AI development

Reduce $\|f_{T_A \to S_A}\|$

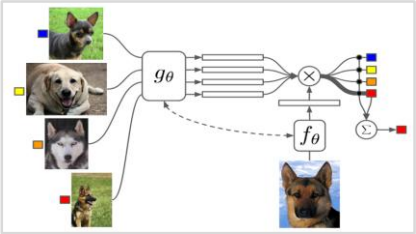
Pre-trained models and NAS



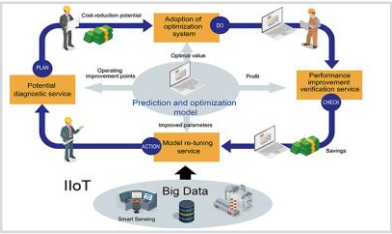
Transfer learning and federated learning



Meta-learning/Few-shot learning



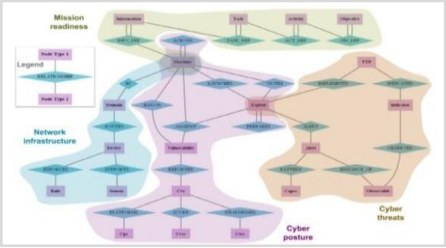
Operations optimization solver



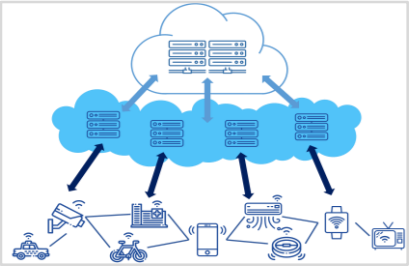
AI implementation

Reduce $\|f_{S_A \to S_{AI}}\|$

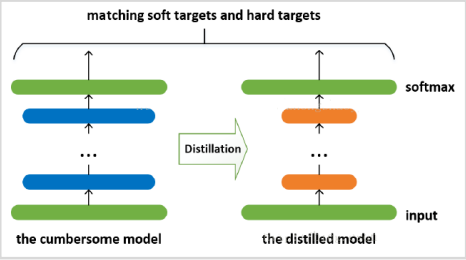
Model explainability and security



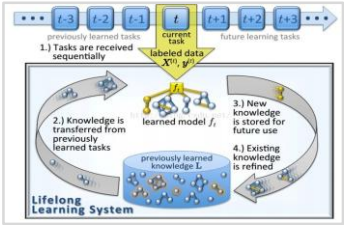
Device-cloud synergy and edge AI



Model compression



Online learning and lifelong learning



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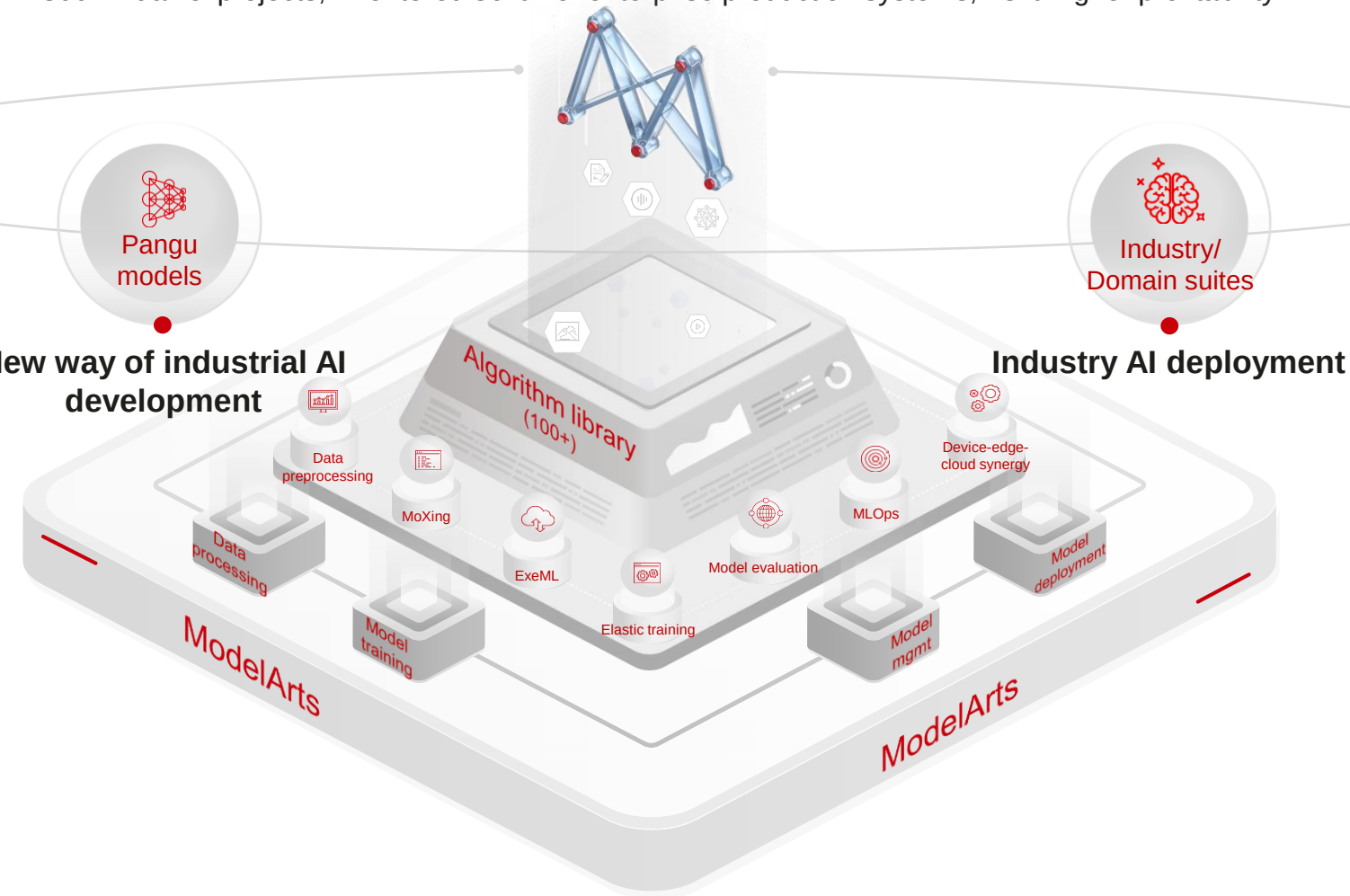
ModelArts: AI Enablement Platform That Drives Intelligent Upgrade Across Industries

City | Transport | Manufacturing | Healthcare | Finance | Network | Scientific research | Meteorology | Water | e-Gov

800+ Huawei projects, AI entered **30%+** of enterprise production systems, **18%** higher profitability



Resource scheduling engine

Unleashed compute power

Pangu models

New way of industrial AI development

Industry/ Domain suites

Industry AI deployment

AI Gallery

Cultivating AI talent and developing an industry ecosystem

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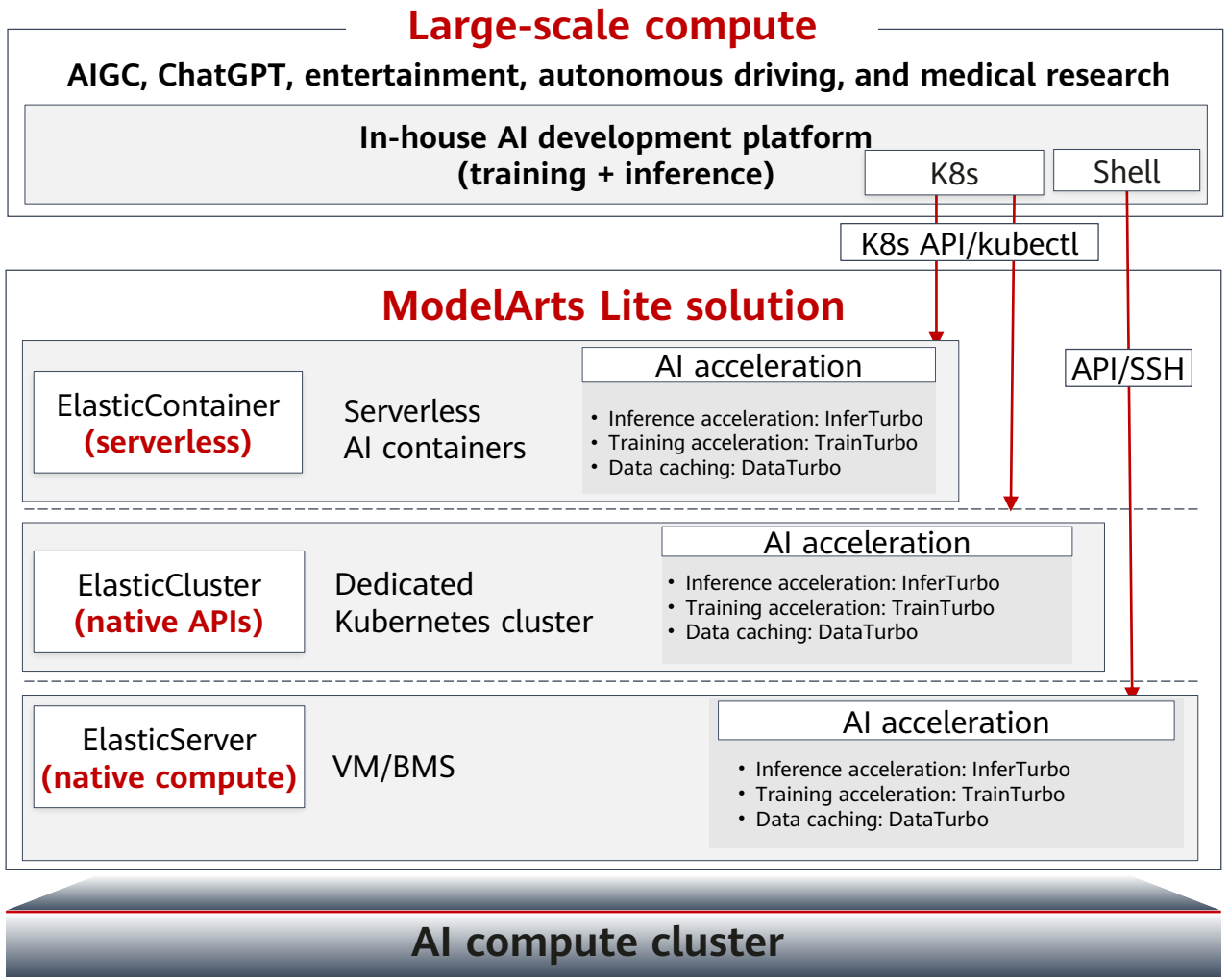
ModelArts Lite: Zero-Cost Model Migration, Adaptation, and Three-Layer Acceleration



Fully compatible with K8s, zero-cost migration and adaptation

- **Serverless NPU containers**
  Auto scaling of NPU containers within seconds
- **Fully compatible with Kubernetes**
  Standard Kubernetes APIs, compatible with popular third-party plugins, zero-cost migration and adaptation

Three-layer acceleration for training and inference on NPUs

- **Inference acceleration: InferTurbo**
  - Heterogeneous multi-core parallel inference: 1.3x AIGC performance.
  - Low-bit compression and heterogeneous operator fusion: 2x inference performance and 1/2 the inference cost for large-scale NLP models.
- **Training acceleration: TrainTurbo**
  Efficient parallel training of large-scale models on 2,048 PUs, 1.8x performance for large-scale NLP models, 1.35x training performance for large-scale autonomous driving models
- **Data caching: DataTurbo**
  PB-scale storage and millisecond-level access latency, 5x data read performance

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ModelArts Empowers RFCx to Conserve Tropical Rainforests

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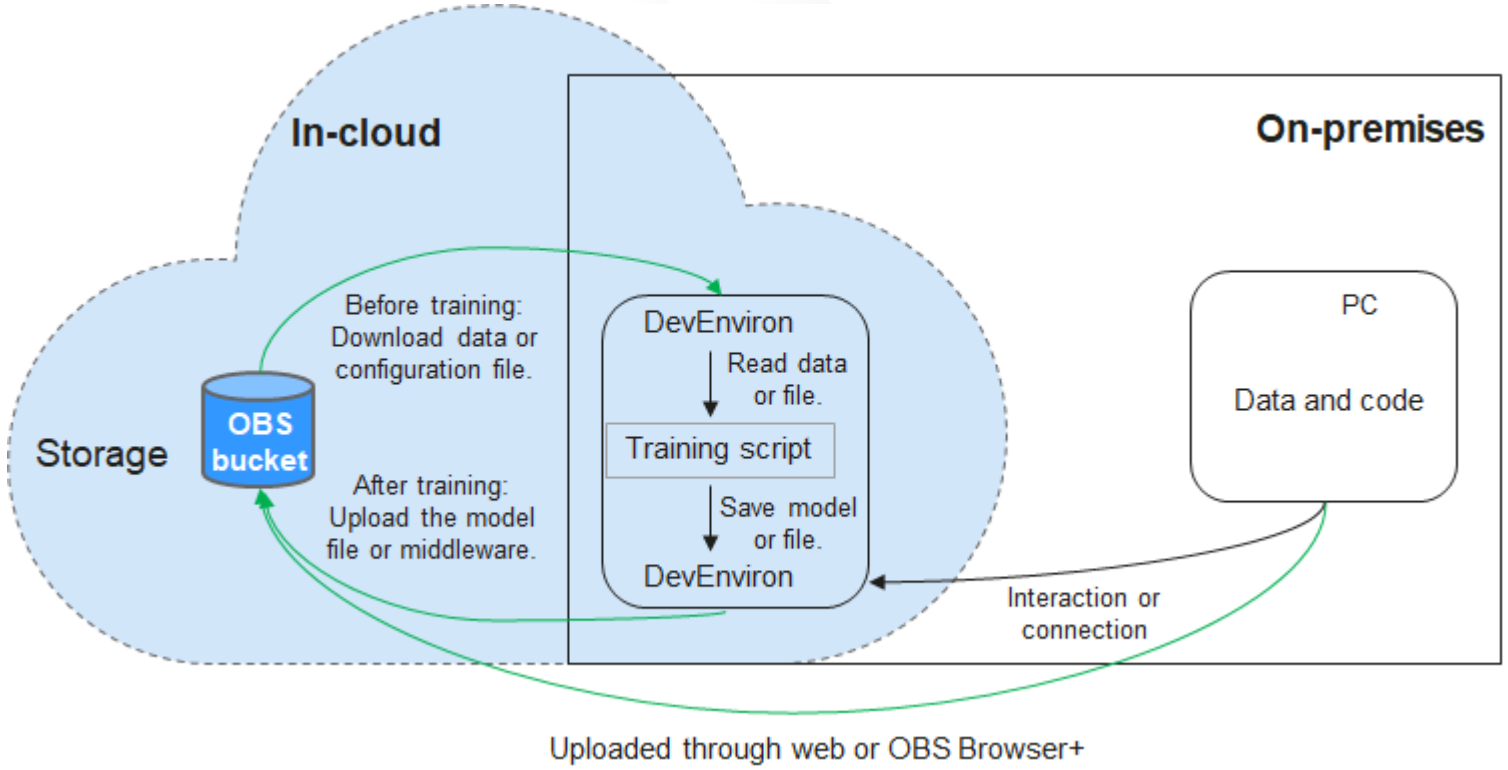
ModelArts notebook for seamless in-cloud and on-premises collaboration

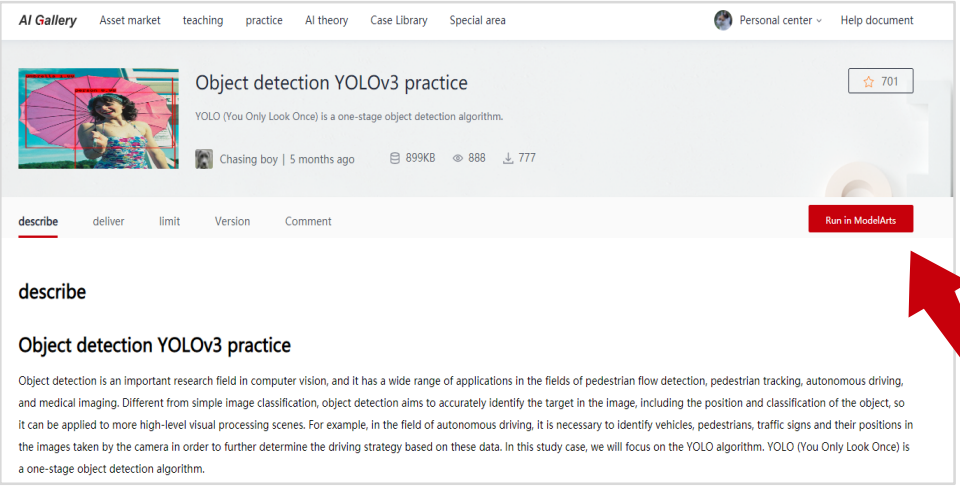


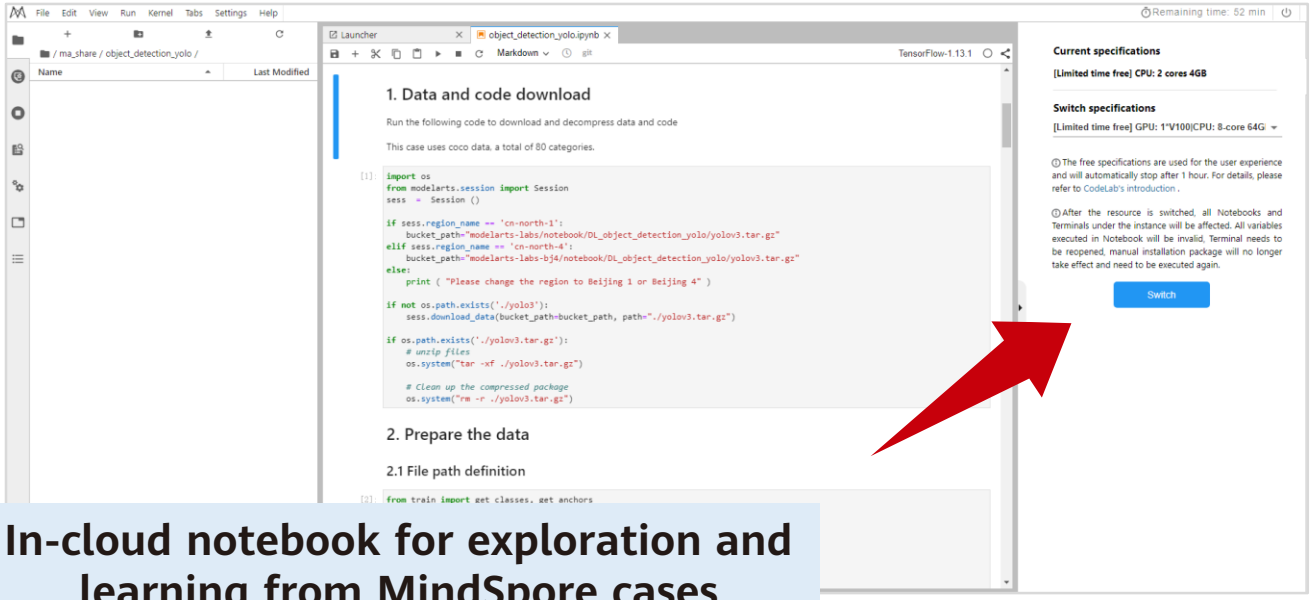
ModelArts CodeLab Makes AI Exploration and Teaching Easier

- In-cloud notebook, case access and sharing in seconds
- Serverless instance management for automated resource reclamation, which is free
- Free compute power and on-demand change of specifications

Keywords

Out-of-the-box usage | Case access | Flexible | Free AI compute power | Sharing and exploration





In-cloud notebook for exploration and learning from MindSpore cases

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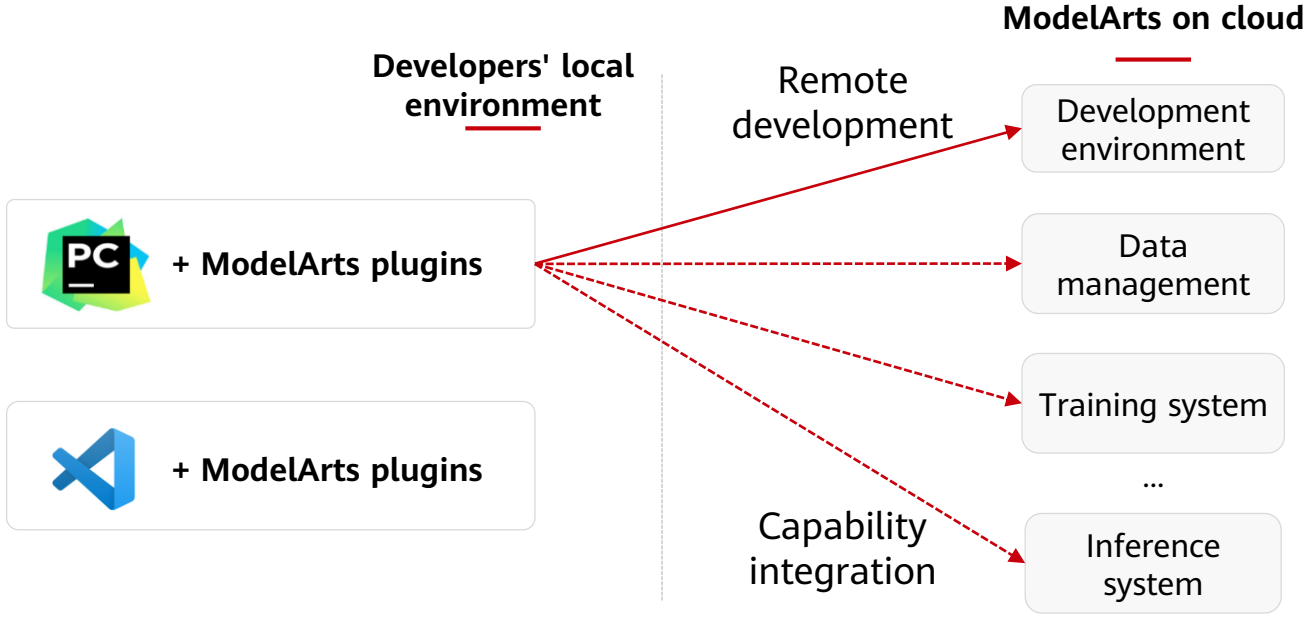
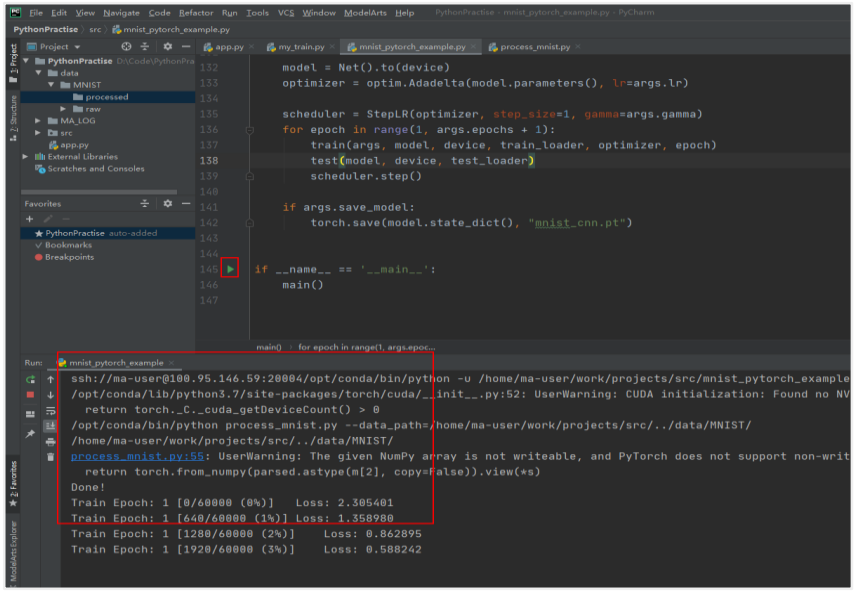
ModelArts IDE SDKs

- Code development and debugging: Local IDE + ModelArts plugins for remote development, tailored to your needs
- Cloud-based development environment with AI compute resources, cloud storage, and built-in AI engines
- Customizable runtime environment: Development environment saved as an image for training and inference

Keywords

ModelArts IDE | Remote development | Multi-person collaboration | Flexible AI compute resources | Efficient development





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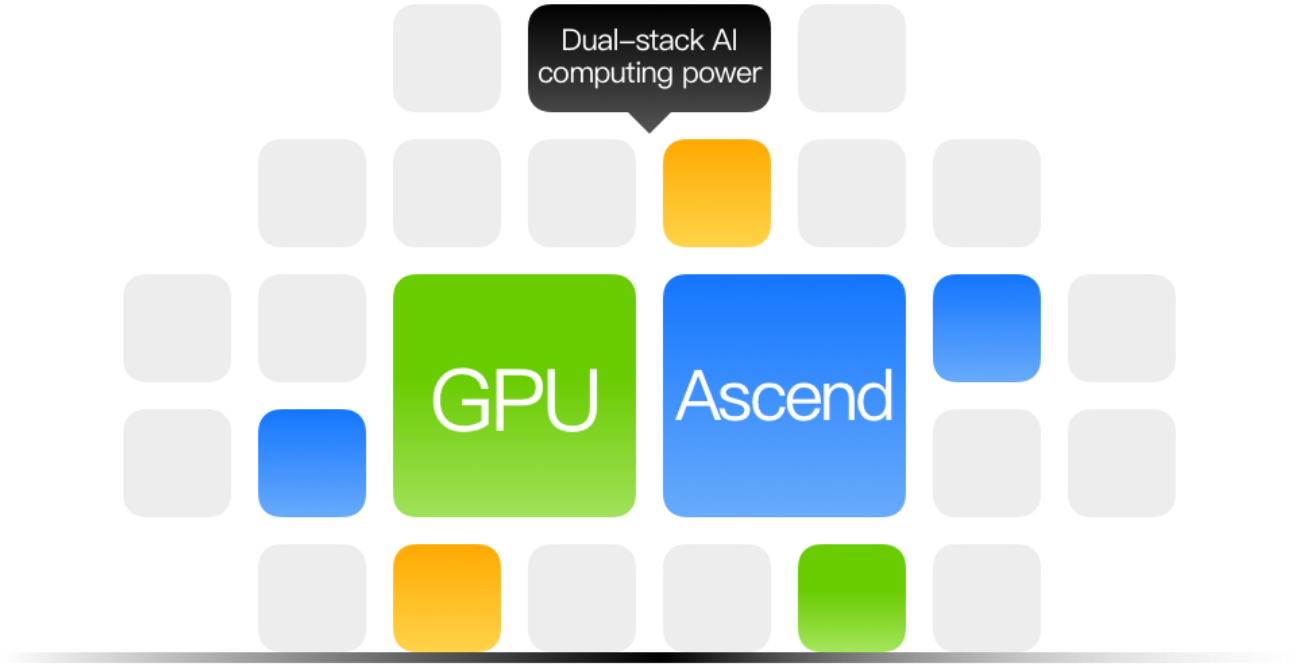
Dual-stack AI computing power: Stable and Secure Computing Base, Fast and Simple Model Training

- GPU and Ascend dual-stack AI computing power supports the management of 10,000-node compute clusters.
- Large-scale distributed training accelerates foundation model development.



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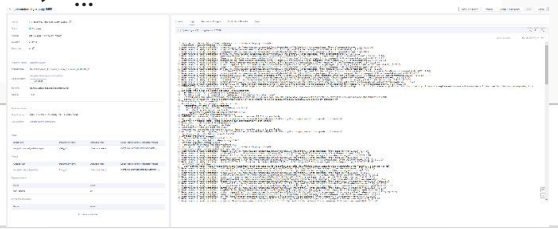
Integrated Development and Training with Intelligent Fault Diagnosis

Algorithm debugging	Before training	In training	After training
IDE-based debugging	FAQs	Job exception prompts	Job failure cause prompts
Trial run	Automatically filtering out abnormal nodes	Automatic fault tolerance and resumable training	Automatic fault classification
✓ Quick job submission using IDE plug-ins ✓ Non-blocking trial run ✓ …	✓ Hardware failures: GPU/CPU/memory/disk/network ✓ Software failures: driver/domain name/system settings	✓ Process I/O check ✓ Resource usage check ✓ Hardware failures: isolation and rectification ✓ Automatic retry upon random faults ✓ Suspension detection ✓ Performance degradation detection ✓ …	✓ Service log analysis ✓ Container return code and event analysis ✓ Analysis of failed tenant tasks ✓ …



Large-scale training: foundation model development, AIGC, autonomous driving, and more

- Large model size, complex logic, and multiple parallel modes
- Large-scale data trained across multiple nodes

① Multiple nodes, long training time, and increased likelihood of errors
② Lack of tools and inefficient fault locating
③ Few monitoring metrics for performance analysis

Small-scale training: industrial AI application development

- Model development using open-source algorithms
- Lack of AI development professionals

① Insufficient problem locating
② Insufficient performance analysis

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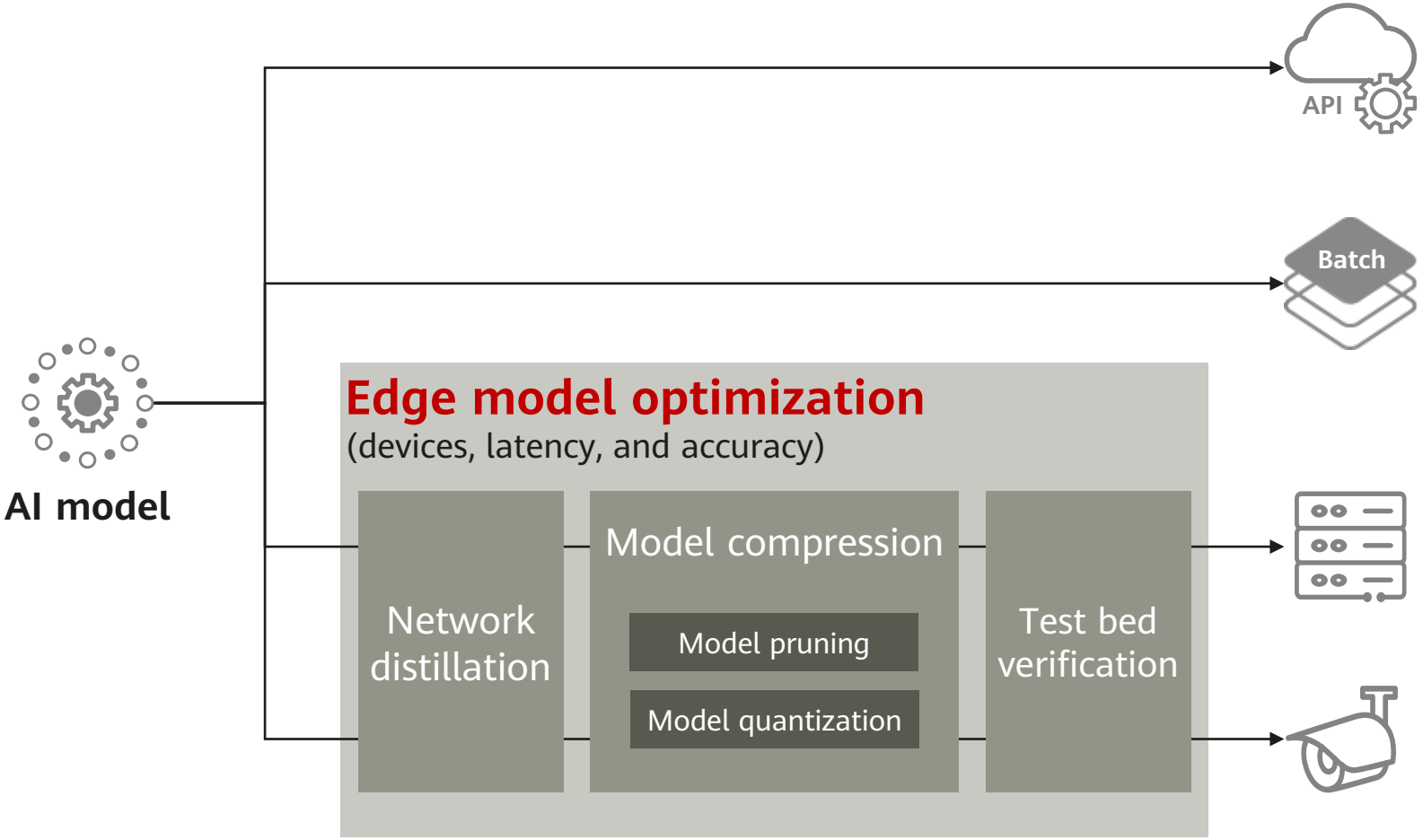
Flexible Model Deployment on Devices, Edge Devices, or the Cloud



Real-time service

- High throughput, low latency, and automatic scale-in
- Inference optimization

Batch service

- Batch data inference task
- Efficient distributed computing

Edge service

- In-depth integration with IEF
- Support for Huawei Ascend AI chips

Edge service

- Support for Huawei HiLens, SDC, and CloudLink

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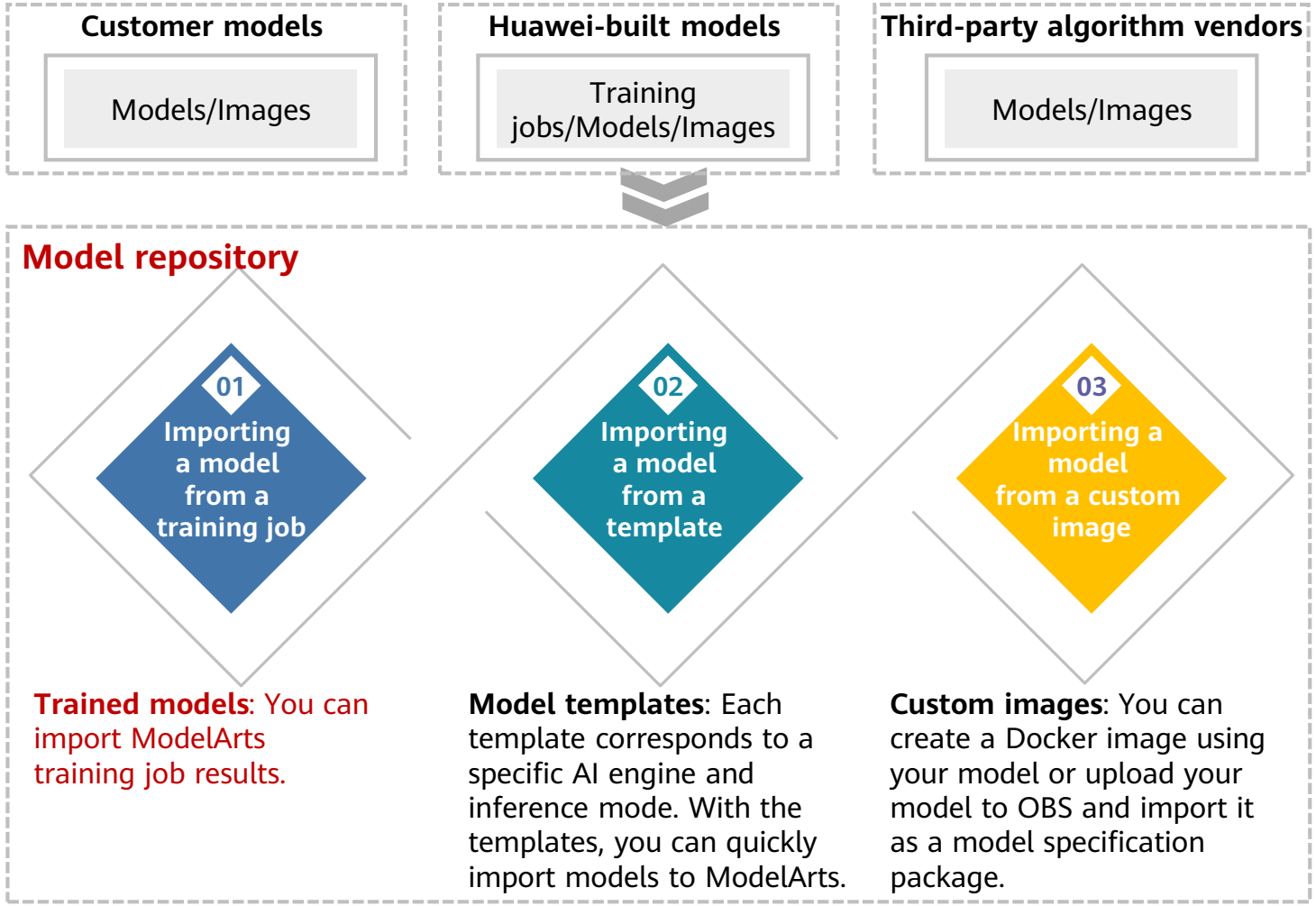
Model Repositories for Unified Management of Models with Different Frameworks and Functions from Different Vendors

Highlights

- AI algorithms with different frameworks and functions from different vendors can be centrally managed.
- Third-party AI algorithms can be quickly released as inference services through Docker images.
- Algorithms can be quickly released as inference services in the form of model files.
- Large model deployment and model update within seconds
- Separate maintenance and upgrade of different AI models
- You can customize the model inference specifications according to your needs. The supported granularity of CPU, GPU, and Ascend 310 is 0.01. The memory allocation has a granularity of 1 MB.
- Easy-to-use operations and management on UI



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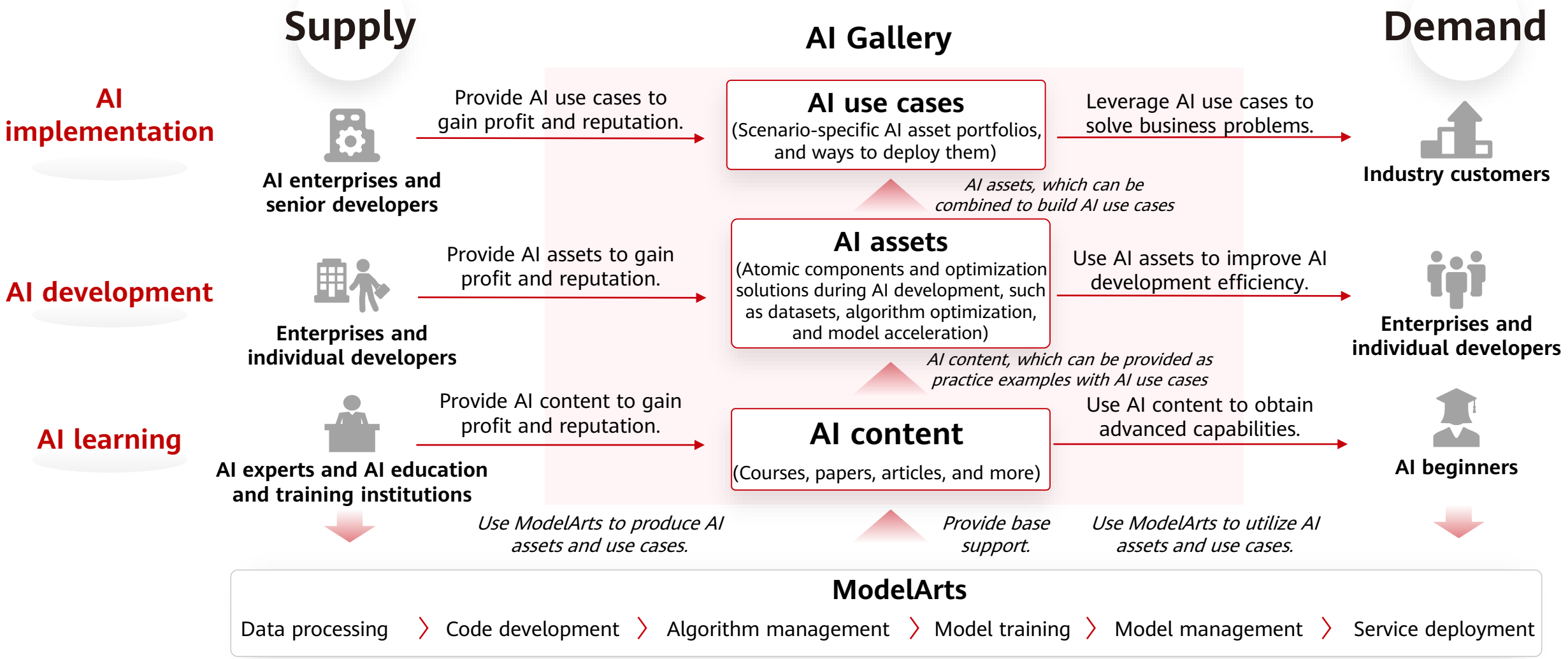
AI Gallery: Bridging Supply and Demand in the AI Ecosystem



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Building a Comprehensive Learning System for AI Developers

Comprehensive courses

AI beginners

- AI concept, structure, and development
- Development and strategic planning of the AI industry
- Huawei's full-stack, all-scenario AI strategies
- AI model creation without writing code

One-stop learning experience

AI engineers

- Python programming knowledge
- Popular frameworks and tools
- MindSpore zone
- Machine learning
- Deep learning
- Domain knowledge application

Follow-up teaching plans

AI application engineers

- Action recognition
- Wildlife identification
- Predictive maintenance
- Face mask detection
- Plant disease detection
- Speech recognition practices
- Lane line detection
- More

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Q&A

1. Which of the following function modules are provided by ModelArts? ( )

A. ExeML

B. Data management

C. Inference platform

D. AI Gallery

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Summary

- This chapter describes the main functions of Huawei one-stop development platform ModelArts, including ExeML, data management, development environment, training, inference, and AI Gallery.

Acronyms and Abbreviations

- Bidirectional Encoder Representations from Transformers (BERT), a pre-trained language representation model
- Recurrent Neural Network (RNN)

Thank You.

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